Jailbreak methods. We consider jailbreak methods that make LLMs accept harmful instructions. We employ three different types of jailbreak methods. (1) Adversarial suffixes (GCG specifically [Zou et al., 2023b]): A sequence of learnable suffix tokens that are optimized to elicit acceptance responses. (2) Persuasion [Zeng et al., 2024]: Persuasion techniques are applied to rephrase naive harmful instructions to persuade LLMs to accept them. (3) Adversarial prompting templates [Yu et al., 2023]: Harmful instructions are inserted into carefully constructed jailbreak prompting templates. Examples of these jailbreak methods are shown in Table 11 in the Appendix.

Refusal rate. Instruct models are usually finetuned to return certain fixed phrases to refuse users' prompts, e.g., ''`Sorry, I cannot`''. To evaluate the models' refusal rate, we follow the convention [Zou et al., 2023b, Arditi et al., 2024, Zhou et al., 2025] to compile a set of common refusal substrings. In Section 3.5, the rate is computed based on the refusal token "No". If the model's response contains one of the refusal substrings, we classify it as refusal; otherwise, it is classified as non-refusal. We calculate the refusal rate out of all the test examples.

3 Decoupling Harmfulness from Refusal

In this section, we investigate the hidden states of harmful/harmless prompts at two different token positions, the last token of the instruction t_{inst} and the last token of the post-instruction tokens $t_{\text{post-inst}}$. This is motivated by our first observation (Section 3.1) that removing all the post-instruction tokens will reduce LLMs' refusal behaviors (Table 1). Next, we demonstrate that harmfulness and refusal may be encoded separately at these two token positions, since the hidden states at t_{inst} form clusters based on the instruction's harmfulness, while the hidden states at $t_{\text{post-inst}}$ form clusters based on whether the instruction is refused (Section 3.2). Then, we quantify the correlation between these harmfulness and refusal clusters, and find they are not always strongly correlated (Section 3.3). Next, we show that steering with the harmfulness direction can also lead to refusal behaviors (Section 3.4). Finally, we show that steering with the harmfulness direction and the refusal direction will lead to opposite behaviors in our designed reply inversion task, providing causal evidence that LLMs encode harmfulness and refusal separately (Section 3.5).

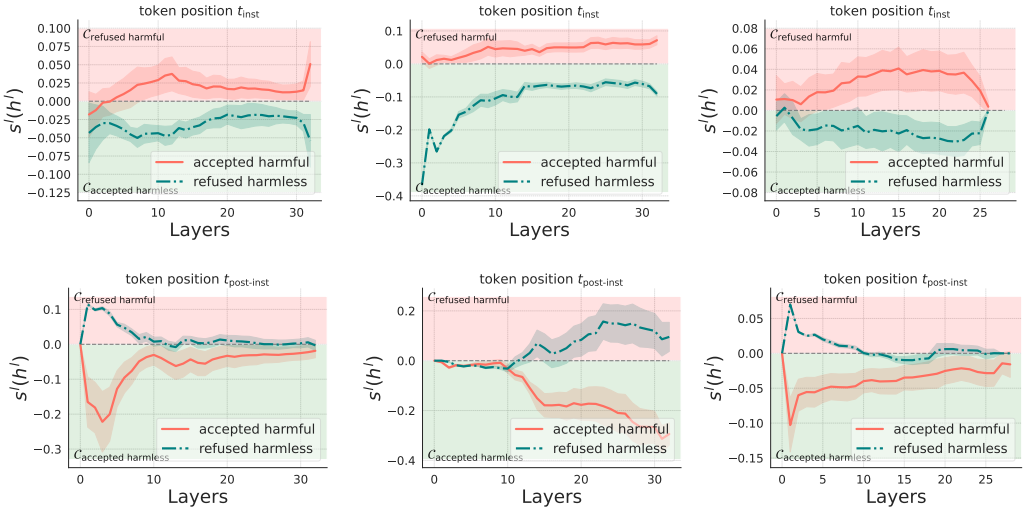
Refusal Rate (%)	w/ post-instruction tokens	w/o post-instruction tokens
LLAMA2-CHAT-7B	100.0	85.3
LLAMA3-INSTRUCT-8B	96.0	58.9
QWEN2-INSTRUCT-7B	98.0	81.3

Table 1: Refusal rates of harmful instructions when prompting with and without post-instruction special tokens in the prompting template. The refusal rate drops dramatically without post-instruction special tokens.

3.1 Removing post-instruction tokens weakens refusal abilities

Observation. We find that LLMs can refuse harmful instructions at $t_{\text{post-inst}}$ while accepting them at t_{inst}. In other words, the refusal ability of harmful instructions can be weakened by removing the post-instruction special tokens in the prompting template. As shown in Table 1, all the tested LLMs have a lower refusal rate of harmful instructions in Advbench [Zou et al., 2023b] when prompted without post-instruction tokens. Examples model outputs are shown in Figure 19 in the Appendix. Past work [Jiang et al., 2025] has shown that different prompting templates can weaken the refusal ability of LLMs. Our results further indicate the importance of post-instruction tokens in generating refusal replies. Those results imply that LLMs may not formulate refusal signals until the post-instruction tokens are passed to the models. Our findings also support the hypothesis of *template-anchored safety alignment* [Leong et al., 2025] that LLMs overly depend on post-instruction tokens in the prompting template to form their decisions of refusal.

Hypothesis. Both $t_{\text{post-inst}}$ and t_{inst} contain the information of the whole input instruction (due to self-attention in Transformers [Vaswani et al., 2017]), but LLMs' refusal behaviors are much stronger



(a) LLAMA3-INSTRUCT-8B (b) LLAMA2-CHAT-7B (c) QWEN2-INSTRUCT-7B

Figure 2: The internal clustering of hidden states extracted at t_{inst} (the first row) and $t_{\text{post-inst}}$ (the second row) exhibit opposing patterns. The red region: $\mathcal{C}^l_{\text{refused harmful}}$ (the cluster of refused harmful instructions). The green region: $\mathcal{C}^l_{\text{accepted harmless}}$ (the cluster of accepted harmless instructions). At each token position, we collect hidden states of two special misbehaving cases: accepted harmful instructions (the red line) and refused harmless instructions (the green line) to see which cluster these two cases fall in. **The first row**: At t_{inst}, across layers, accepted harmful instructions (the red line) mostly fall in $\mathcal{C}^l_{\text{refused harmful}}$ (the red region), while refused harmless instructions (the green line) mostly fall in $\mathcal{C}^l_{\text{accepted harmless}}$ (the green region). This implies that the clustering may be based on whether the instruction is harmful or harmless, regardless of whether it is refused or accepted. **The second row**: At $t_{\text{post-inst}}$, the clustering behavior is **reversed**. Now the accepted harmful instructions (the red line) fall in $\mathcal{C}^l_{\text{accepted harmless}}$ (the green region), while refused harmless instructions (the green line) fall in $\mathcal{C}^l_{\text{refused harmful}}$ (the red region). This implies that at $t_{\text{post-inst}}$, the clustering may be based on whether the instruction is refused or accepted, regardless of whether it is harmful or harmless. In Section 3.5, we further provide causal evidence supporting that harmfulness/ harmlessness features are encoded at t_{inst}, while refusal/acceptance features are encoded at $t_{\text{post-inst}}$.

at $t_{\text{post-inst}}$. We then ask: What is encoded at t_{inst}? Is that different from the refusal signals encoded at $t_{\text{post-inst}}$? We hypothesize that: at t_{inst}, the hidden states of harmful instructions may encode harmfulness, and then at $t_{\text{post-inst}}$, the hidden states will encode explicit refusal signals for the model to generate the rejection responses. We verify our hypothesis in Section 3.2 by analyzing how the hidden states of different instructions (harmful but accepted, and harmless but rejected) form clusters at different token positions.

3.2 Hidden states cluster by harmfulness at t_{inst}, and by refusal at $t_{\text{post-inst}}$

Motivated by the different refusal behaviors with and without $t_{\text{post-inst}}$, we extract hidden states at t_{inst} and $t_{\text{post-inst}}$ to examine what each position encodes. As hidden states often form distinct clusters based on the input features they encode [Zheng et al., 2024, Marks and Tegmark, 2023, Tigges et al., 2023], we analyze how harmful/ harmless instructions that lead to different models' behaviors form clusters at t_{inst} and $t_{\text{post-inst}}$. Specifically, we ask: Is the clustering in the latent space based on,(1) the instruction's harmfulness/ harmlessness or (2) its refusal/acceptance? To investigate this question, we first compute the clusters of hidden states for instructions with desired model behaviors (refused harmful instructions and accepted harmless instructions). We then analyze the misbehaving instructions (*accepted but harmful* instructions, and *refused but harmless* instructions) to see which cluster they fall in. For instance, if the hidden states of accepted but harmful instructions are closer to the cluster of refused harmful instructions than that of accepted harmless instructions, it suggests that the instruction's harmfulness/harmlessness plays a more important role in the clustering than its refusal/acceptance.

Instruction clustering. We first collect the hidden states of accepted harmful instructions, refused harmful instructions, accepted harmless instructions, and refused harmless instructions at t_{inst} and $t_{\text{post-inst}}$ respectively (data used are detailed in Section 2). Then, at each layer l, we compute the cluster of refused harmful instructions ($\mathcal{C}^l_{\text{refused harmful}}$), and the cluster of accepted harmless instructions ($\mathcal{C}^l_{\text{accepted harmless}}$) at the studied token position on the training set. The cluster centers are the mean of these instructions' hidden states and are denoted as $\mu^l_{\text{harmful refused}}$ and $\mu^l_{\text{harmless accepted}}$. To decide which cluster a test instruction x belongs to at each layer l, we calculate the cosine similarity between its hidden states h^l and the two cluster centers, $cos_sim(h^l, \mu^l_{\text{refused harmful}})$ and $cos_sim(h^l, \mu^l_{\text{accepted harmless}})$. Then we calculate the following:

$$s^l(h^l) = cos_sim(h^l, \mu^l_{\text{refused harmful}}) - cos_sim(h^l, \mu^l_{\text{accepted harmless}}). \tag{2}$$

If $s^l(h^l) > a$, $h^l \in \mathcal{C}^l_{\text{refused harmful}}$; If $s^l(h^l) < a$, then $h^l \in \mathcal{C}^l_{\text{accepted harmless}}$. We by default set the threshold a as 0 in this work, which has an intuitive mathematical interpretation: h^l is assigned to the cluster whose center it is closer to. However, the oracle value for a in LLMs may not necessarily be 0, as internal clusters are likely to be fuzzy and overlapped. We leave further investigation on estimating the oracle as future work. We then compute the average $s^l(h^l)$ for all the misbehaving accepted harmful instructions and refused harmless instructions at each token position to see, on average, which cluster these misbehaving examples are closer to. This allows us to assess whether clustering is primarily driven by the refusal/acceptance feature or the harmful/harmless feature.

At t_{inst}, hidden states primarily form clusters by harmfulness; at $t_{\text{post-inst}}$, hidden states form cluster by refusal. The results of different models are shown in Figure 2. We find that at t_{inst}, *harmfulness* plays a more decisive role in clustering, while at $t_{\text{post-inst}}$, *refusal* plays a more important role in clustering. For example, at t_{inst} (the first row of Figure 2), for all three models tested, across all layers, the hidden states of accepted harmful instructions (the red solid line) mainly fall in the $\mathcal{C}^l_{\text{refused harmful}}$ cluster, and refused harmless instructions (the green dashed line) mainly fall in the $\mathcal{C}^l_{\text{accepted harmless}}$ cluster (the green region). These results suggest that at t_{inst}, the clustering is driven more by the harmfulness feature of the instructions than by whether they were refused. However, at $t_{\text{post-inst}}$ (the second row of Figure 2), the clustering behavior is reversed. The hidden states of refused harmless instructions (the green dashed line) fall in $\mathcal{C}^l_{\text{refused harmful}}$ (the red region), and the accepted harmful instructions (the red solid line) fall in $\mathcal{C}^l_{\text{accepted harmless}}$ (the green region). These results suggest that the clustering is driven more by whether the instruction was accepted or refused, regardless of whether it was actually harmful or harmless. Apart from the two positions t_{inst} and $t_{\text{post-inst}}$, we also study the clustering patterns at more token positions and perform similar layer-wise analysis in Appendix F.1. We find that clustering based on the harmfulness of instructions is the most evident at t_{inst}.

Additionally, in both Llama3 and Qwen2 models, we observe a similar layer-wise pattern that at $t_{\text{post-inst}}$, refused harmless instructions (the green dashed line) start to fall in $\mathcal{C}^l_{\text{refused harmful}}$ (the red region) and accepted harmful instructions (the red solid line) in $\mathcal{C}^l_{\text{accepted harmless}}$ (the green region) in early layers, indicating the early emergence of strong refusal signals. However, such strong refusal signals do not appear until the later layers for Llama2. We suspect this is due to Llama2's limited capabilities, requiring deeper layers to make refusal decisions. We leave it as future work to understand the role of each layer. Overall, our results demonstrate at t_{inst} and $t_{\text{post-inst}}$, the internal clustering of instructions exhibit opposing patterns, which suggests that harmfulness may be encoded at t_{inst}, while refusal may be encoded at $t_{\text{post-inst}}$.

3.3 Correlation between beliefs of harmfulness and refusal

In this section, we quantitatively analyze the correlation between the belief of harmfulness and the belief of refusal. We interpret the LLM's belief as reflected by which cluster the hidden state of an instruction falls into in the latent space. **We find that sometimes the model may internally recognize the correct level of harmfulness in input instructions, yet still exhibit incorrect refusal or acceptance behaviors.** Formally, following the clustering analysis in Section 3.2, we define the cluster formed by harmful instructions at position t_{inst} as the harmfulness cluster $\mathcal{C}^l_{\text{harmful}}$ in layer l, whose center is denoted as $\mu^{l,\,t_{\text{inst}}}_{\text{harmful}}$. Similarly, we denote the harmlessness cluster at t_{inst} as $\mathcal{C}^l_{\text{harmless}}$ and its center as $\mu^{l,\,t_{\text{inst}}}_{\text{harmless}}$. Then at $t_{\text{post-inst}}$, we denote the clusters formed by refused and accepted instructions as $\mathcal{C}^l_{\text{refusal}}$ and $\mathcal{C}^l_{\text{accept}}$ respectively, whose centers are $\mu^{l,\,t_{\text{inst}}}_{\text{refusal}}$ and $\mu^{l,\,t_{\text{inst}}}_{\text{accept}}$. For an input